

Continuum Mechanics HW2

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Technion

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1 Vorticity

1. Write the definition of ω_{ij} and multiply both sides by ε_{ijl} . Simplify to obtain a single summand.
2. Multiply the original definition by ε_{ijl} and use Levi-Civita tensor identities to obtain

$$\omega = \nabla \times \mathbf{v} \quad (1)$$

1.1 part 1

The definition of the vorticity is

$$\omega_{ij} = -\frac{1}{2}\varepsilon_{ijk}\omega_k \quad (2)$$

multiplying by ε_{ijl}

$$\varepsilon_{ijl}\omega_{ij} = -\frac{1}{2}\varepsilon_{ijl}\varepsilon_{ijk}\omega_k \quad (3)$$

$$\varepsilon_{ijl}\omega_{ij} = -\frac{1}{2}\left(2\delta_{kl} \cdot (-1)^2\right)\omega_k \quad (4)$$

$$\varepsilon_{ijl}\omega_{ij} = -\omega_l \quad (5)$$

1.2 part 2

By original definition I imagine this is what's meant:

$$\omega_{ij} = \frac{1}{2}\left(\frac{\partial v_i}{\partial x_j} - \frac{\partial v_j}{\partial x_i}\right) \quad (6)$$

multiplying both sides by ε_{ijl}

$$\varepsilon_{ijl}\omega_{ij} = \frac{1}{2}\varepsilon_{ijl}\left(\frac{\partial v_i}{\partial x_j} - \frac{\partial v_j}{\partial x_i}\right) \quad (7)$$

$$= \frac{1}{2}\left(\varepsilon_{ijl}\frac{\partial v_i}{\partial x_j} - \varepsilon_{ijl}\frac{\partial v_j}{\partial x_i}\right) \quad (8)$$

$$= -\varepsilon_{ijl}\frac{\partial v_i}{\partial x_j} \quad (9)$$

combining with the result of part 1 we get

$$\omega_l = \varepsilon_{ijl}\frac{\partial v_i}{\partial x_j} \quad (10)$$

which by definition is

$$\omega = \nabla \times \mathbf{v} \quad (11)$$

2 Streamlines and pathlines

Consider the following 2D flow

$$\mathbf{v}(x, y) = \begin{pmatrix} \alpha y t \\ \beta \end{pmatrix} \quad (12)$$

1. Find the streamlines $\mathbf{x}(x, t)$ and the pathlines $\mathbf{x}(t, \mathbf{x}_0)$ of the flow.
2. Create an animation of the motion of a particle starting at position $(0, 0)$ at $t = 0$ and $\alpha = \beta = 1$. In the animation plot you should display the streamlines at different times, as well as the full trajectory of the particle. Discuss the relation between the streamlines and the trajectory of the particle.

2.1 Streamlines and Pathlines

As we know the streamlines satisfy

$$\frac{dx_s}{\alpha y t} = \frac{dy_s}{\beta} \quad (13)$$

therefore

$$\frac{dx_s}{dy_s} = \frac{\alpha y t}{\beta} \quad (14)$$

integrating w.r.t y

$$x_s = \frac{\alpha t}{2\beta} y^2 + C \quad (15)$$

so the streamlines are

$$x_s - \frac{\alpha t}{2\beta} y^2 = C \quad (16)$$

2.2 Animation

Animation is attached in "trajectory_animation.mp4". The trajectory the particle took is such that it moved along the streamline at each point in time as it was then. i.e the streamline at each moment in time determined the direction the particle followed for that step.

3 Free Flow

1. Find the eulerian and lagrangian velocities of a non accelerated flow ($\frac{D}{Dt}v = 0$) in one dimension, given the initial condition

$$v(t=0) = -ax_0, \quad a > 0 \quad (17)$$

What is the PDE solved by the eulerian velocity? Show that your solution satisfies it.

2. What happens to the velocity at time $t = a^{-1}$? Explain both from the eulerian and lagrangian perspectives.

3.1 non-accelerated flow

For the lagrangian perspective we'll look at the lagrangian velocity at the initial condition

$$\dot{x}(x_0, t=0) = -ax_0 \implies x(x_0, t) = -ax_0t + C \quad (18)$$

I'm assuming I'm to treat x_0 as the initial condition for the position, therefore

$$x(x_0, 0) = C = x_0 \quad (19)$$

so

$$x(x_0, t) = -ax_0t + x_0 = x_0(1 - at) \quad (20)$$

this lets us move to the eulerian picture by looking at the initial position in reference to the rest of the frame

$$x_0 = \frac{x}{1 - at} \quad (21)$$

inserting this back into the lagrangian velocity

$$v_E = \frac{-ax}{1 - at} \quad (22)$$

This solves the PDE

$$\partial_t v + v \partial_x v = 0 \quad (23)$$

we can show this by inserting the solution

$$\partial_t \left(\frac{-ax}{1 - at} \right) + \left(\frac{-ax}{1 - at} \right) \partial_x \left(\frac{-ax}{1 - at} \right) = 0 \quad (24)$$

simplifying

$$\frac{-a^2x}{(1 - at)^2} + \left(\frac{-ax}{1 - at} \right) \left(\frac{-a}{1 - at} \right) = 0 \quad (25)$$

$$\frac{-a^2x}{(1 - at)^2} + \frac{a^2x}{(1 - at)^2} = 0 \quad (26)$$

$$0 = 0 \quad (27)$$

3.2 at that time

plugging in $t = a^{-1}$ into the position of the particle we get

$$x(x_0, a^{-1}) = x_0 (q - a \cdot a^{-1}) = 0 \quad (28)$$

so in the lagrangian perspective the particle will be at position $x = 0$. In the eulerian perspective we can plug in the time and find

$$v_E = \frac{-ax}{1 - a \cdot a^{-1}} = \text{undefined} \quad (29)$$

If we combine the information from both perspectives, that the flow is entirely at $x = 0$ and that the velocity is undefined, that there needs to be an infinite velocity of the fluid to compress it all into the origin.

4 Poiseuille Flow Between Concentric Cylinders

Assume that a fluid flows between two infinite concentric cylinders with radii ($r_2 > r_1$), which are at rest. The flow is parallel to the symmetry axis $\hat{\mathbf{z}}$. The fluid is Newtonian with viscosity μ (satisfying the Navier-Stokes equations). Assume there are no external forces and that the pressure gradient is constant and also directed along the symmetry axis: ($\partial_z p = \text{const.}$)

1. Write the full equation of motion in cylindrical coordinates.
2. Use the symmetries of the problem to simplify the equation. What is the form of the nonlinear term $(\mathbf{v} \cdot \nabla) \mathbf{v}$?
3. Write the boundary conditions of the problem and solve the resulting system to find the velocity profile.
4. Define and compute the average velocity through a cross-sectional area, $\bar{U}$. Use the average velocity to define the Reynolds number Re and the momentum flux through a cross-section. Explain the meaning of these two quantities.
5. Compute the friction force (per unit area) acting on the cylinder walls and express it in dimensionless form using the momentum flux you defined. What is the dependence of the result on Re ? Explain this dependence.
6. Use your solutions from parts 3-5 to find the flow profile and the friction force in a hollow cylinder.

4.1 Equations of motion

Since the fluid is Newtonian, the NS equation relevant to us is

$$\rho (\partial_t + (\mathbf{v} \cdot \nabla)) \mathbf{v} = -\nabla p + \mu \nabla^2 \mathbf{v} + \rho \mathbf{f} \quad (30)$$

inserting the definitions of the differential operators in cylindrical

$$\rho \left(\partial_t + \left(\mathbf{v} \cdot \left(\frac{\partial}{\partial r} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial}{\partial \theta} \hat{\boldsymbol{\theta}} + \frac{\partial}{\partial z} \hat{\mathbf{z}} \right) \right) \right) \mathbf{v} = - \left(\frac{\partial}{\partial r} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial}{\partial \theta} \hat{\boldsymbol{\theta}} + \frac{\partial}{\partial z} \hat{\mathbf{z}} \right) p + \mu \left(\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} + \frac{\partial^2}{\partial z^2} \right) \mathbf{v} + \rho \mathbf{f} \quad (31)$$

which is awful.

4.2 simplification

Since we assume the flow is parallel to $\hat{\mathbf{z}}$, there is no external force, and the pressure gradient is constant, we can simplify to

$$\rho (\partial_t + v_z \partial_z) \mathbf{v} = -\partial_z p + \mu \nabla^2 \mathbf{v} \quad (32)$$

To simplify the laplacian we can note that the velocity clearly can't have dependence on the radial direction or the z position, from symmetry, therefore

$$\rho (\partial_t) \mathbf{v} = -\partial_z p + \mu \left(\frac{1}{r} \partial_r (r \partial_r) \right) \mathbf{v} \quad (33)$$

From symmetry and conservation laws, the flow must remain constant in time, therefore

$$0 = -\partial_z p + \mu \left(\frac{1}{r} \partial_r (r \partial_r) \right) \mathbf{v} \quad (34)$$

rearranging

$$\partial_z p = \mu \left(\frac{1}{r} \partial_r (r \partial_r) \right) \mathbf{v} \quad (35)$$

4.3 Boundary Conditions

Since both cylinders are stationary we must demand that

$$v(r_1) = v(r_2) = 0 \quad (36)$$

We'll rearrange the equation to be for the velocity

$$\left(\frac{1}{r} \partial_r (r \partial_r) \right) \mathbf{v} = \frac{\partial_z p}{\mu} \quad (37)$$

$$(\partial_r (r \partial_r)) \mathbf{v} = \frac{r \partial_z p}{\mu} \quad (38)$$

$$r \partial_r v = \frac{\partial_z p}{2\mu} r^2 + B \quad (39)$$

$$v = \frac{\partial_z p}{4\mu} r^2 + B \ln(r) + C \quad (40)$$

using the boundary conditions

$$\frac{\partial_z p}{4\mu} r_1^2 + B \ln(r_1) + C = \frac{\partial_z p}{4\mu} r_2^2 + B \ln(r_2) + C = 0 \quad (41)$$

$$B \ln(r_1) - B \ln(r_2) = \frac{\partial_z p}{4\mu} r_2^2 - \frac{\partial_z p}{4\mu} r_1^2 \quad (42)$$

$$B \ln\left(\frac{r_1}{r_2}\right) = \frac{\partial_z p}{4\mu} (r_2^2 - r_1^2) \quad (43)$$

$$B = \frac{\partial_z p (r_2^2 - r_1^2)}{4\mu \ln\left(\frac{r_1}{r_2}\right)} \quad (44)$$

inserting back into one of the equations

$$\frac{\partial_z p}{4\mu} r_1^2 + \frac{\partial_z p (r_2^2 - r_1^2)}{4\mu \ln\left(\frac{r_1}{r_2}\right)} \ln(r_1) + C = 0 \quad (45)$$

$$\frac{\partial_z p}{4\mu} \left(r_1^2 + \frac{(r_2^2 - r_1^2)}{\ln\left(\frac{r_1}{r_2}\right)} \ln(r_1) \right) + C = 0 \quad (46)$$

$$C = -\frac{\partial_z p}{4\mu} \left(r_1^2 + \frac{(r_2^2 - r_1^2)}{\ln\left(\frac{r_1}{r_2}\right)} \ln(r_1) \right) \quad (47)$$

4.4 average velocity

The first way to do the average that comes to mind is

$$\bar{U} = \frac{1}{A} \int_A v \, dS \quad (48)$$

inputting the expressions we have and integrating in polar coordinates

$$\bar{U} = \frac{1}{\pi(r_2^2 - r_1^2)} \int_A dS \frac{\partial_z p}{4\mu} r^2 + B \ln(r) + C \quad (49)$$

$$= \frac{2}{(r_2^2 - r_1^2)} \int_{r_1}^{r_2} dr \, r \left(\frac{\partial_z p}{4\mu} r^2 + B \ln(r) + C \right) \quad (50)$$

$$= \frac{2}{(r_2^2 - r_1^2)} \left(\frac{\partial_z p}{16\mu} (r_2^4 - r_1^4) + B \left(\frac{r_2^2 \ln(r_2) - r_1^2 \ln(r_1)}{2} + \frac{r_1^2 - r_2^2}{4} \right) + \frac{C}{2} (r_2^2 - r_1^2) \right) \quad (51)$$

$$= \frac{\partial_z p (r_1^2 + r_2^2)}{8\mu} + B \left(\frac{r_2^2 \ln(r_2) - r_1^2 \ln(r_1)}{r_2^2 - r_1^2} - \frac{1}{2} \right) + C \quad (52)$$

We can define the Reynolds number as

$$Re = \frac{\rho \bar{U} l}{\mu} \quad (53)$$

where l is the characteristic length of the system, which since we only have one such quantity

$$l = r_2 - r_1 \quad (54)$$

so

$$Re = \frac{\rho \bar{U} (r_2 - r_1)}{\mu} \quad (55)$$

and we can define the momentum flux as

$$\Phi = \int_A dS \, \rho \bar{U} = 2\pi\rho \int_A dr \, r \bar{U} \quad (56)$$

which I'll leave like this

4.5 Friction Force

We can use the shear stress to find the friction

$$\sigma_{rz} = -\mu \frac{\partial v_z}{\partial r} = -\mu \left(\frac{\partial_z p}{2\mu} r + \frac{B}{r} \right) = -\frac{\mu B}{r} - \frac{\partial_z p}{2} r \quad (57)$$

and a force element is

$$dF = \sigma_{rz} \hat{\mathbf{r}} \, dA \quad (58)$$

so

$$\frac{dF}{dA} = -\frac{\mu B}{r} - \frac{\partial_z p}{2} r \quad (59)$$

force over area has units of pressure. Momentum flux also has units of pressure. Therefore we can define a dimensionless form by

$$\tilde{F} = \frac{\frac{dF}{dA}}{\Phi} \quad (60)$$

and the total force is

$$\tilde{F}(r_2) - \tilde{F}(r_1) = \frac{-1}{\Phi} \left(\frac{\mu B}{r_2} + \frac{\partial_z p}{2} r_2 \right) + \frac{1}{\Phi} \left(\frac{\mu B}{r_1} + \frac{\partial_z p}{2} r_1 \right) \quad (61)$$

$$= \frac{1}{\Phi} \left(\mu B \left(\frac{1}{r_1} - \frac{1}{r_2} \right) - \frac{\partial_z p}{2} (r_1 - r_2) \right) \quad (62)$$

$$= \frac{1}{\Phi} \left(\frac{\mu B}{r_1 r_2} - \frac{\partial_z p}{2} \right) (r_2 - r_1) \quad (63)$$

$$= \frac{\mu}{\rho \bar{U} \Phi} \left(\frac{\mu B}{r_1 r_2} - \frac{\partial_z p}{2} \right) Re \quad (64)$$

4.6 Hollow Cylinder

Since the cylinder is hollow, we have no boundary condition from the inner cylinder so we can set $B = 0$ giving us

$$v = \frac{\partial_z p}{4\mu} + C \quad (65)$$

giving us the force per area

$$\frac{dF}{dA} = -\frac{\partial_z p}{2} r_2 \quad (66)$$

5 Flow over inclined plane

Consider a layer of a viscous (also newtonian, isotropic, and incompressible) fluid with thickness h bounded below by an inclined plane at an angle α to the horizon. We look for a steady state.

1. Using the symmetries of the problem, reduce $\mathbf{v}(t, x, y, z)$ to a single variable, single component field.
2. Write N-S equations in explicit form (with indices), and solve them to get general solutions for v and p . Leave the constants that arise.
3. Assume the atmospheric pressure is p_0 . Use this and non-slip condition to express the needed components of the stress tensor on the boundaries.
4. Find the constants and write the full solutions.

5.1 Symmetries

First of all, since we're looking for a steady state, it would not depend on the time. Since the fluid is isotropic, it does not depend on x or y , giving us $\mathbf{v}(z)$. We can further reduce this by using the fact the fluid is incompressible, giving us $\nabla \cdot \mathbf{v} = 0$. Due to symmetry this leaves

$$\mathbf{v}(z) = v_x \hat{\mathbf{x}} \quad (67)$$

5.2 Navier-Stokes

N-S is

$$\rho (\partial_t + \mathbf{v} \cdot \nabla) \mathbf{v} = -\nabla p + \mu \nabla^2 \mathbf{v} + \rho \mathbf{f} \quad (68)$$

where the force $\mathbf{f}$ is gravitation

$$\mathbf{f} = -g (\sin \alpha \hat{\mathbf{x}} + \cos \alpha \hat{\mathbf{z}}) \quad (69)$$

moving to index notation

$$\rho (\partial_t + v^j \partial_j) v^i = -\partial_i p + \mu \partial_j \partial_j v^i - \rho g (\sin \alpha \delta_1^i + \cos \alpha \delta_3^i) \quad (70)$$

This can be decomposed into an equation for each spatial direction

$$\rho (\partial_t + v^j \partial_j) v^x = -\partial_x p + \mu \partial_j \partial_j v^x - \rho g (\sin \alpha \delta_1^x + \cos \alpha \delta_3^x) \quad (71)$$

$$\rho (\partial_t + v^j \partial_j) v^y = -\partial_y p + \mu \partial_j \partial_j v^y - \rho g (\sin \alpha \delta_1^y + \cos \alpha \delta_3^y) \quad (72)$$

$$\rho (\partial_t + v^j \partial_j) v^z = -\partial_z p + \mu \partial_j \partial_j v^z - \rho g (\sin \alpha \delta_1^z + \cos \alpha \delta_3^z) \quad (73)$$

simplifying to

$$0 = -\partial_x p + \mu \partial_z \partial_z v^x - \rho g \sin \alpha \quad (74)$$

$$0 = -\partial_y p \quad (75)$$

$$0 = -\partial_z p - \rho g \cos \alpha \quad (76)$$

therefore

$$p_z = -\rho g z \cos \alpha \quad (77)$$

$$p_y = \text{const.} \quad (78)$$

$$p_x = -\rho g z \sin \alpha + \mu \partial_z v^x \quad (79)$$

and v_x using the first equation

$$\frac{\partial^2 v}{\partial z^2} = \frac{1}{\mu} (\partial_x p + \rho g \sin \alpha) \quad (80)$$

therefore

$$v = \frac{1}{2\mu} (\partial_x p + \rho g \sin \alpha) z^2 + Az + B \quad (81)$$

but one of the symmetries we found earlier is that the velocity has no dependence on the x coordinate so $\partial_x p$ must be a constant (which it is from what we found earlier, and it's even 0)

$$v = \frac{\rho g \sin \alpha}{2\mu} z^2 + Az + B \quad (82)$$

5.3 boundaries

The no slip condition and atmospheric pressure conditions give us

$$v(z=0) = 0 \quad \sigma_{xz} = 0 \quad p(z=h) = p_0 \quad (83)$$

which in turn gives us

$$B = 0 \quad A = \frac{\rho g h \sin \alpha}{\mu} \quad (84)$$

and I couldn't finish this due to time constraints.